Module 1: Basic Linear Algebra

1. Matrices

- Definition: A matrix is a tabular representation of data organized into rows and columns.
- Square Matrix: A matrix where the number of rows equals the number of columns.
- Identity Matrix: A square matrix with 1s on the main diagonal and 0s elsewhere.
- Zero Matrix: A matrix in which all elements are zero.
- Matrix Representation ($Ax = b$): A system of linear equations can be written as $Ax = b$, where A is the coefficient matrix, x is the vector of unknowns, and b is the constant vector.

2. Systems of Linear Equations

- Homogeneous System: A system where the constant vector b is zero ($Ax = 0$).
- Non-homogeneous System: A system where $b \neq 0$.
- Trivial Solution: For $Ax = 0$, the solution $x = 0$ is called the trivial solution.
- Consistency: A system is consistent if it has at least one solution.

3. Linear Independence

- Definition: Vectors are linearly independent if c1v1 + c2v2 + ... + cnvn = 0 has only the trivial solution.
- Linear Dependence: If there exists a non-trivial solution, the vectors are linearly dependent.
- Theorem: If the number of vectors exceeds the dimension of the space, the vectors are linearly dependent.

4. Vector Spaces

- Definition: A vector space is a set of vectors closed under addition and scalar multiplication.
- R^n Space: Elements are n-tuples of real numbers.
- Dimension: The number of vectors in a basis. Example: dim(R^3) = 3.

5. Basis

- Definition: A set of vectors is a basis if it is linearly independent and spans the vector space.
- Standard Basis of R^3: {(1,0,0), (0,1,0), (0,0,1)}.

6. Rank of a Matrix

- Meaning: Rank is the number of linearly independent rows or columns.
- Property: $\text{Rank}(A) \leq \min(m, n)$ for an $m \times n$ matrix.
- Row-Echelon Form: Number of non-zero rows equals the rank.

7. Linear Transformations

- Definition: A linear transformation preserves addition and scalar multiplication.
- Kernel (Null Space): Set of vectors x such that $T(x) = 0$.
- Image (Column Space): Set of all possible outputs of T.